

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTECPE405A - MICROCONTROLLER AND ADVANCED PROCESSOR

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 What is the size of the Program Counter (PC) in 8051?
a) 8-bit b) 10-bit c) 12-bit d) 16-bit
- 2 In 8051, mode 2 of Timer works as:
a) 13-bit timer b) 16-bit timer c) Auto-reload d) Counter only
- 3 How many interrupt sources are available in 8051?
a) 3 b) 5 c) 6 d) 8
- 4 Which interrupt is used for serial data reception in 8051?
a) INT0 b) INT1 c) RI d) TI
- 5 8051 microcontroller has how many 16 bit registers?
a) 2 b) 3 c) 1 d) 0
- 6 In LCD interfacing, which pin is used to select between command and data register?
a) RS b) RW c) EN d) VSS
- 7 The ADC0808 has how many analog input channels?
a) 4 b) 6 c) 8 d) 10
- 8 Which of the following is not a mode of operation in ARM7?
a) Supervisor b) Abort c) Monitor d) Undefined
- 9 In ARM 7 architecture, which register stores the return address after a subroutine call?
a) SP b) R15 c) LR d) PC
- 10 The LPC2148 microcontroller is based on which processor core?
a) ARM Cortex-M4 b) ARM7TDMI-S c) ARM9E d) Cortex-M3
- 11 LPC 2148 pro development board has _____ on chip memory.
a) 500k b) 625k c) 512k d) 425k
- 12 LPC2148 supports how many 32-bit timers?
a) 1 b) 2 c) 4 d) 0

Q2. Solve the following.

- A) Explain the differences between Harvard and Von Neumann architectures.

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B) Describe memory organization of the 8051 microcontroller. 6

Q3. Solve the following.

A) Write an 8051 assembly language program to generate a square wave of frequency 6 10 KHz on pin P1.0. Assume that XTAL = 11.0592 MHz.

B) Draw an interfacing diagram to interface a 7 segment display to 8051 6 microcontroller and write an ALP to display '0'.

Q4. Solve Any Two of the following.

A) Draw an interfacing diagram to interface a relay to 8051 microcontroller and 6 write an ALP to turn it ON and OFF with certain time delay.

B) Write an 8051 assembly program to rotate a stepper motor clockwise using Port 2. 6

C) Describe the process of interfacing DAC0808 with 8051 and write an ALP to 6 display a triangular waveform.

Q5. Solve Any Two of the following.

A) Explain significance of each bit of flag register of ARM 7 processor. 6

B) Discuss all the modes of operation of ARM 7 processor. 6

C) Describe ARM7 processor data flow model. 6

Q6. Solve Any Two of the following.

A) Explain memory map of LPC2148 microcontroller. 6

B) Draw an architecture of LPC2148 microcontroller and list out its features. 6

C) Write a note on renesas microcontroller RL78/G13. 6

*** End ***